Decoding Wrist EMG to Text with Recurrent, Convolutional-Recurrent, and Transformer Architectures

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Abstract

This project investigates the problem of decoding typed text from wrist electromyography (EMG) signals. Starting from the provided EMG-to-QWERTY baseline codebase, several alternative sequence modeling architectures were evaluated on a single-user dataset. The baseline time-depth separable convolution (TDS) model was compared against a bidirectional LSTM, a bidirectional GRU, a ResNet18 + LSTM architecture, and a Transformer encoder trained with CTC loss. All models used the same preprocessing pipeline based on log-spectrogram features, data augmentation, and greedy CTC decoding.

Among the evaluated architectures, the bidirectional LSTM achieved the best overall performance, reducing test character error rate (CER) from 24.18% for the baseline to 18.95%. The GRU achieved the lowest validation CER at 18.14%, but generalized poorly to the held-out test session due to a large increase in insertion errors, yielding a test CER of 26.09%. The ResNet18 + LSTM model was significantly more computationally expensive while underperforming the simpler recurrent models. The Transformer encoder improved steadily after a long warm-up phase and reached 24.04% validation CER, but full-session test evaluation failed because the sinusoidal positional encoding was limited to sequences of length 5000, far shorter than the held-out test session.

Overall, the results suggest that in this single-user EMG typing setting, relatively simple recurrent architectures, particularly bidirectional LSTMs, provide the best trade-off between accuracy, robustness, and training efficiency.

1 Introduction

This project investigates how different sequence modeling architectures affect performance on the EMG-to-QWERTY decoding dataset. Starting from the provided time-depth separable convolution (TDS) baseline, the objective was to determine whether models with stronger temporal modeling capacity could reduce character error rate (CER) on the single-user dataset.

Although the baseline convolutional model performs reasonably well, EMG typing signals exhibit complex temporal structure that may be better captured by architectures designed explicitly for sequential data. Recurrent neural networks such as LSTMs and GRUs provide direct mechanisms for modeling temporal dependencies, while convolutional–recurrent hybrids combine spatial feature extraction with sequence modeling. Transformer-based architectures offer an alternative approach by modeling long-range relationships through self-attention.

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To explore these design choices, several post-convolution architectures were implemented and evaluated under a shared preprocessing and training pipeline. Specifically, the baseline TDS model was compared against a bidirectional LSTM, a bidirectional GRU, a ResNet18 + LSTM hybrid, and a Transformer encoder trained with CTC loss. All models used the same input representation, augmentation pipeline, and decoding procedure to ensure fair comparison.

The primary goal of this work was to evaluate how these architectural choices influence decoding accuracy, training behavior, and generalization to a held-out test session. Performance was measured using character error rate together with insertion, deletion, and substitution error analysis. The results show that relatively simple recurrent architectures, particularly the bidirectional LSTM, provide the strongest and most consistent performance in this single-user EMG typing setting.

2 Methods

2.1 Dataset and split

Experiments were conducted on the provided single-user EMG typing dataset. The data consist of multiple recording sessions from a single participant wearing EMG recording bands on both wrists. The official train, validation, and test splits provided with the project codebase were used.

During training and validation, windowed segments were sampled from their respective sessions. At test time, the full held-out session was decoded as a single continuous sequence to better approximate realistic typing conditions.

2.2 Input representation

Each example consisted of EMG recordings sampled at 2000 Hz from two recording bands, each containing 16 electrode channels, for a total of 32 channels. Raw EMG signals were converted into log-spectrogram features using a short-time Fourier transform with the following parameters:

- FFT size: 64
- hop length: 16

This produced a time-frequency representation for each electrode channel. The resulting tensors had shape $(T, N, 2, 16, F)$, where T denotes temporal length, N is batch size, and F is the number of frequency bins.

2.3 Feature preprocessing

All architectures shared the same frontend feature processing pipeline. First, spectrogram inputs were normalized using batch normalization applied independently to each electrode channel. Next, a multi-band rotation-invariant MLP processed each EMG band separately while pooling across small electrode rotations. This design improves robustness to small shifts in electrode placement while preserving the underlying muscle activity patterns.

The resulting per-band features were concatenated and flattened before being passed to the sequence encoder.

2.4 Data augmentation

All experiments used the same preprocessing and augmentation pipeline during training:

- conversion to tensor
- random band rotation with offsets $\{-1, 0, 1\}$
- temporal alignment jitter with maximum offset 120
- log-spectrogram transformation
- SpecAugment with time masking and frequency masking

Validation and test data were processed deterministically without augmentation.

2.5 Training setup

All models were trained using the same optimization configuration:

- optimizer: Adam
- learning rate: 0.001
- scheduler: linear warmup with cosine annealing
- batch size: 32
- training duration: up to 40 epochs

Models were trained using the Connectionist Temporal Classification (CTC) loss. During evaluation, predictions were generated using greedy CTC decoding.

Model performance was evaluated using character error rate (CER). To better analyze error patterns, insertion error rate (IER), deletion error rate (DER), and substitution error rate (SER) were also reported.

2.6 Architectures

All models share the same preprocessing and feature extraction pipeline described above. The primary difference between architectures lies in the sequence encoder used to model temporal structure in the EMG signals.

2.6.1 TDS baseline

The provided baseline model uses a time-depth separable convolution (TDS) encoder applied to the extracted spectrogram features. The encoder consists of stacked temporal convolution blocks with residual connections that model temporal dependencies without explicit recurrence. This architecture serves as the convolutional reference point for comparison with alternative sequence models.

2.6.2 Bidirectional LSTM

To evaluate recurrent sequence modeling, the convolutional encoder was replaced with a stacked bidirectional LSTM. The model used three LSTM layers with hidden size 512 and dropout of 0.1 between layers. Bidirectional recurrence allows the model to incorporate both past and future temporal context when predicting characters. Because the LSTM preserves temporal resolution, its outputs are directly compatible with the CTC training framework.

2.6.3 Bidirectional GRU

A bidirectional GRU architecture was also evaluated using the same dimensional configuration as the LSTM. The GRU replaces the LSTM memory cell with a simpler gating mechanism, resulting in fewer parameters while maintaining similar sequence modeling capacity. This model provides a more computationally efficient recurrent alternative.

2.6.4 ResNet18 + LSTM

To explore whether stronger convolutional feature extraction improves performance, a hybrid architecture consisting of a ResNet18 encoder followed by a bidirectional LSTM was implemented. The ResNet18 backbone processes each spectrogram frame independently and produces a compact spatial feature representation. These features are then fed into the LSTM, which models temporal dependencies across frames.

2.6.5 Transformer encoder

Finally, a Transformer encoder was implemented to evaluate an attention-based sequence modeling approach. The model first projects the input features to a fixed embedding dimension and adds sinusoidal positional encoding to preserve temporal order. A stack of multi-head self-attention layers then models long-range temporal relationships across the sequence. The Transformer was trained with the same CTC objective used for the other architectures.

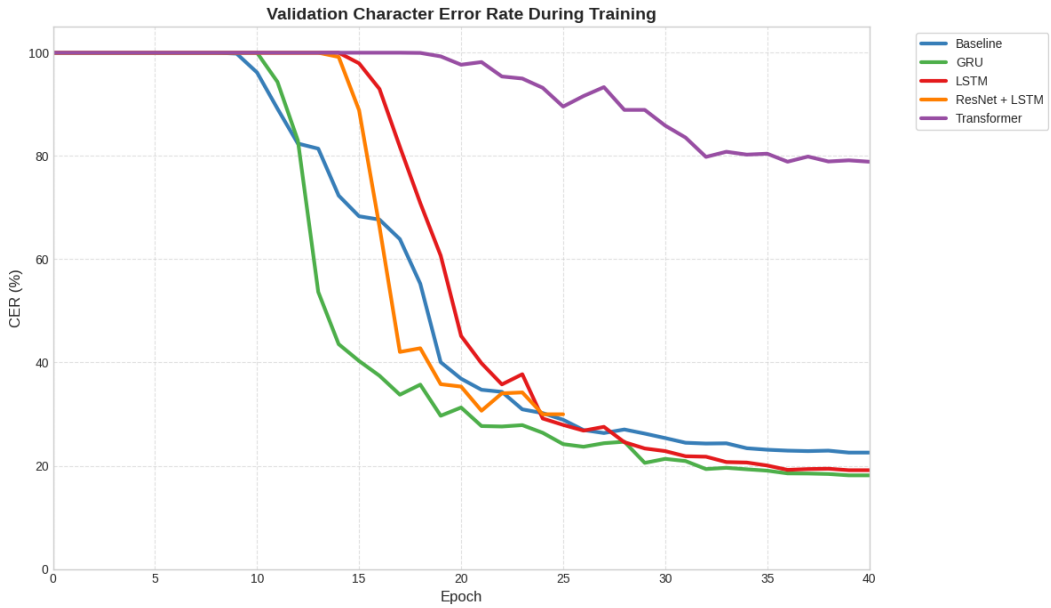

Figure 1: Validation CER during training for the baseline, GRU, LSTM, ResNet18 + LSTM, and Transformer over the first 40 epochs.

3 Results

3.1 Final quantitative comparison

Table 1 summarizes the final performance of all evaluated architectures. Performance is reported in terms of character error rate (CER), with lower values indicating better decoding accuracy.

Table 1: Model comparison

Model	Parameters	Best Val CER (%)	Test CER (%)
TDS baseline	5.3M	22.55	24.18
Bidirectional LSTM	18.4M	19.14	18.95
Bidirectional GRU	13.9M	18.14	26.09
ResNet18 + LSTM	21.8M	29.97	28.42
Transformer	3.8M	24.04	N/A

The bidirectional LSTM achieved the best overall performance, reducing test CER from 24.18% for the baseline model to 18.95%. This represents the largest improvement among all tested architectures.

The GRU achieved the lowest validation CER (18.14%) but exhibited substantially worse performance on the held-out test session, suggesting weaker generalization. The ResNet18 + LSTM hybrid performed worse than both recurrent models and the baseline despite having the largest parameter count. The Transformer reached a validation CER of 24.04%, slightly worse than the baseline, but full-session test evaluation could not be completed because the sinusoidal positional encoding was limited to sequences of length 5000, whereas the held-out test session had length 140757.

3.2 Training dynamics

Figure 1 shows validation CER as a function of training epoch for all models during the first 40 epochs.

Across all architectures, validation CER remained extremely high during the early stages of training. During the first several epochs the models produced long random character sequences under the CTC objective, resulting in very large insertion counts and CER values far exceeding 100%. For clarity,

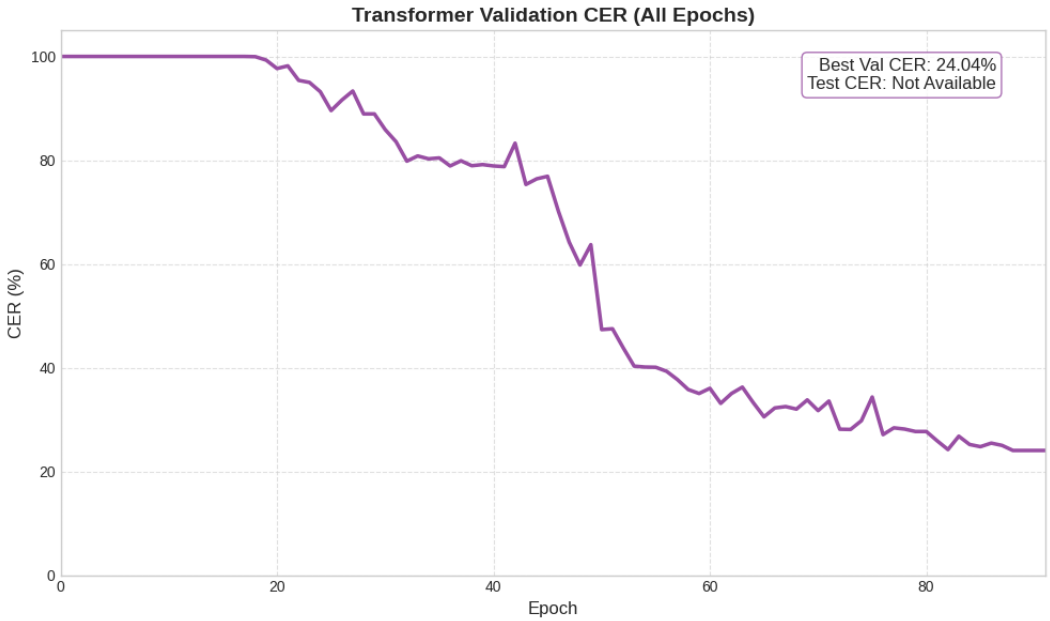

Figure 2: Transformer validation CER over all epochs.

this initial region is not shown in the figure. Meaningful learning began after approximately 10–15 epochs, when the models started producing mostly blank predictions and gradually learned correct temporal alignments.

Once learning began, the GRU exhibited the fastest early improvement, with validation CER dropping sharply shortly after epoch 12. The baseline TDS model also improved quickly and showed stable convergence throughout training.

The ResNet18 + LSTM hybrid improved more slowly than the baseline and converged to a higher final validation error than the GRU. The bidirectional LSTM reduced error more gradually during the early stages of training but eventually reached one of the lowest validation CER values.

In contrast, the Transformer displayed a much longer warm-up period and improved more slowly throughout the first 40 epochs. Its validation CER remained substantially higher than the recurrent models during this training window.

3.3 Transformer long-run behavior

To better understand the Transformer training behavior, the model was trained for a longer duration beyond the initial 40 epochs. The resulting validation CER trajectory is shown in Figure 2.

The Transformer required a significantly longer training horizon before producing meaningful predictions. Validation CER remained close to 100% for the first several epochs before gradually decreasing as training progressed. After approximately 50 epochs the model began to improve more rapidly, eventually reaching a best validation CER of 24.04%.

Although this represents a large improvement relative to its early training behavior, the final performance remained worse than the recurrent models.

3.4 Error breakdown analysis

To better understand model behavior, the test CER was decomposed into insertion, deletion, and substitution error rates for several architectures.

For the best LSTM checkpoint:

- test CER: 18.95

- test IER: 2.77
- test DER: 2.57
- test SER: 13.62

For the GRU:

- test CER: 26.09
- test IER: 12.25
- test DER: 0.48
- test SER: 13.36

For the baseline:

- test CER: 24.18
- test IER: 6.27
- test DER: 2.14
- test SER: 15.78

Comparing the LSTM to the baseline shows that the largest improvement comes from a substantial reduction in insertion errors (6.27% to 2.77%). Substitution errors are also reduced moderately (15.78% to 13.62%), while deletion errors increase slightly (2.14% to 2.57%). Overall, the reduction in insertion errors accounts for the majority of the CER improvement, with additional gains from fewer substitutions.

In contrast, the GRU produces a very large number of insertion errors (12.25%) while maintaining a very low deletion rate (0.48%). This indicates that the model produces relatively few deletion errors but frequently predicts additional characters that are not present in the ground truth. Because substitution rates are similar between the GRU and LSTM (13.36% vs 13.62%), the higher CER of the GRU is primarily explained by its substantially larger insertion error rate.

These results suggest that the GRU tends to emit characters too aggressively during decoding, producing many spurious outputs. In contrast, the LSTM exhibits a more balanced error profile with substantially fewer insertion errors, which ultimately leads to better overall CER.

4 Discussion

These experiments suggest that recurrent sequence models are particularly well suited for single-user EMG typing. Among all evaluated architectures, the bidirectional LSTM achieved the best overall performance, reducing test CER from 24.18% for the baseline to 18.95%. In contrast, the GRU achieved the lowest validation CER but generalized poorly to the held-out test session, indicating that strong validation performance alone did not guarantee robust full-session decoding.

One likely explanation for the difference between the LSTM and GRU is their error profile under CTC decoding. The GRU converged more quickly during training, but this came with a substantial increase in insertion errors at test time. Its very low deletion rate suggests that the model rarely omitted true characters, but instead emitted too many additional characters that were not present in the ground truth. The LSTM, in contrast, exhibited a more balanced trade-off between insertion, deletion, and substitution errors, which led to lower overall CER. In this setting, stable decoding behavior appeared more important than achieving the absolute lowest validation CER.

The ResNet18 + LSTM hybrid did not provide the expected benefits of stronger convolutional feature extraction. In practice, this architecture learned much more slowly than the other models and produced worse validation and test performance despite having the largest parameter count. One possible explanation is that the ResNet18 frontend was too heavy for this single-user setting, making optimization unnecessarily difficult relative to the amount of available data. A smaller convolutional frontend may preserve the potential benefits of local feature extraction while reducing training cost and improving optimization stability.

The Transformer encoder showed a different failure mode. Although its validation CER improved steadily with longer training, full-session test evaluation could not be completed because the sinusoidal positional encoding was constructed with a maximum length of 5000, whereas the held-out test session had length 140757. As a result, the model could be validated on windowed examples but could not be applied to the full continuous test sequence used for final evaluation. This means the Transformer results are incomplete: the observed validation performance shows that the model was learning, but its true full-session test performance could not be measured without modifying the implementation. Increasing the maximum positional encoding length or replacing it with a more scalable positional representation would be necessary for a fair comparison.

Overall, these results indicate that relatively simple recurrent architectures remain highly effective for EMG-based sequence decoding when data are limited and long continuous sequences must be decoded reliably. While more complex architectures such as convolutional-recurrent hybrids and Transformers may offer advantages in principle, they were harder to optimize and did not outperform the bidirectional LSTM in this setting.

5 Limitations

Several limitations should be considered when interpreting the results of this study.

First, all experiments were conducted on a single-user dataset. EMG signals are highly user-specific due to differences in muscle physiology, electrode placement, and typing behavior. As a result, models that perform well for one user may not generalize well to other individuals. Evaluating the architectures on a multi-user dataset would be necessary to determine whether the observed performance differences remain consistent across users.

Second, the Transformer evaluation was incomplete due to a positional encoding limitation in the implementation. The sinusoidal positional encoding was initialized with a maximum sequence length of 5000, while the full held-out test session contained over 140000 time steps. This prevented the model from being evaluated on the full continuous test sequence. Although validation performance suggests that the Transformer was learning meaningful representations, its true test performance remains unknown without modifying the positional encoding implementation.

Third, hyperparameter tuning was limited. All models were trained using largely shared optimization settings to ensure fair comparisons and reduce experimentation time. However, some architectures, particularly the Transformer and the convolutional-recurrent hybrid, may require different learning rates, training schedules, or regularization strategies to reach their best performance.

Fourth, computational constraints limited the scope of architectural exploration. The ResNet18 + LSTM model was substantially more expensive to train than the other models and converged slowly, which restricted the number of experiments that could be performed. Exploring smaller convolutional frontends or alternative hybrid architectures may yield better performance without the optimization difficulties observed with the ResNet18 backbone.

Finally, decoding was performed using greedy CTC decoding for all models. More advanced decoding strategies, such as beam search or language model integration, could potentially improve character error rates by correcting some insertion and substitution errors. These approaches were not explored in this work.

6 Conclusion

This project compared several sequence modeling architectures for decoding typed text from wrist EMG signals using a shared preprocessing and training pipeline. Among the evaluated models, the bidirectional LSTM achieved the best overall performance, reducing test CER from 24.18% to 18.95%. Although the GRU achieved the lowest validation CER, it generalized poorly due to a large increase in insertion errors. More complex architectures, including the ResNet18 + LSTM hybrid and the Transformer encoder, did not outperform the simpler recurrent model in this single-user setting. Overall, the results suggest that bidirectional LSTMs provide a strong and reliable baseline for EMG-based text decoding.

References

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